# SMART PATIO SHIELD

## Management Report

*Predicting wind-drive-rain as a service*
*The opportunity, the evidence, and the path to the product*

*The wet veranda that started it: wind-driven-rain blowing through the grillwork, exactly the event this system predicts.*


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## Executive Summary

Smart Patio Shield is a machine learning system that predicts, hour by hour and location by location, when wind-driven rain is about to affect a specific outdoor space, early enough to trigger protective action before the rain arrives. The problem it addresses is ordinary but costly: a covered but open-sided space stays dry in normal rain, yet rain driven sideways by gusting winds blow straight in, and no one can watch the sky all day to act in time.

Built as a Master's capstone using five years of Jamaican weather data and satellite imagery, is system is a complete, validated working prototype. All three planned models are built and evaluated on data they never saw during the development. The best model predicts wind-driven-rain events far better than chance and substantially better than conventional rule-of-thumb approaches. The project also tested whether adding satellite imagery improved the predictions and found that it did not, so the deployed system runs on lightweight weather data alone. That makes it cheaper to run and simpler to license than a more complex system would be.

The same prediction capability has a clear commercial path as a licensable service for any organisation exposed to sudden, localised rain. Promising stakeholders include hotels and restaurants protecting outdoor spaces; agricultural operations timing weather-sensitive work and protecting sun-drying crops; government agencies in tourism, agriculture and entertainment; and manufacturers of automated screens and awnings seeking an intelligent control layer. Because the system is built entirely on free public data and low-cost computing, the marginal cost of extending it to a new location is low. That is a structural commercial advantage.

This report sets out the problem and opportunity, what has been built and proven, the value proposition and the stakeholders it could serve, an honest account of the risks and limitations, and the path from prototype to product. The commercial opportunity is presented as a hypothesis: the demand is plausible and specific, but not yet validated with paying customers, and the report is explicit about that distinction throughout.

**AT A GLANCE**

**What it does:** predicts wind-driven rain events for a specific location, hour by hour.

**Status:** complete, validated prototype. All three models built and evaluated; deployable service demonstrated.

**Key finding:** strong prediction using free weather data alone; satellite imagery was tested and found to add no value at this scale, keeping the product lightweight.

**Commercial form:** a licensable prediction service (API), priced per site, with low onboarding cost.

**Best-fit stakeholders:** tourism, agriculture, entertainment, hospitality, and automated-screen manufacturers.

# 1. The problem and the Opportunity

## a. An everyday, costly problem

Across Jamaica, and the wider Caribbean, a great deal of valuable activity happens in spaces that are covered overhead but open at the sides: verandas, patios, restaurant terraces, hotel dining areas, poolside lounges, event marquees, and agricultural drying yards. These spaces shed ordinarily vertical rain. What they cannot shed is rain driven sideways by gusting wind, which passes straight through open sides and grillwork and reaches everything inside.

The cost is rarely catastrophic but is frequent and cumulative: damaged furnishings and stock, interrupted dining and events, spoiled produce, and a degraded customer experience. The underlying difficulty is one of timing. Nobody can watch the sky continuously, and by the time wind-driven rain is obviously arriving, it is already too late.

## b. Why existing tools do not solve it

General weather forecasts do not address this problem, for two reasons. First, they are not specific enough: a regional "40% chance of rain" says nothing about whether a particular terrace, in the next hour, will be hit by the combination of rain and wind that actually cause damage. Second, they predict the wrong thing: the event that matters here is not rainfall alone but the joint occurrence of rain and gusting wind at a specific spot. No mainstream service predicts that compound, hyper-local event.

## c. The opportunity

The opportunity is to provide exactly what is missing: a prediction that is specific to a location, specific to the wind-driven-rain event, and early enough to act on. Delivered as a service, this becomes a single capability that many different organisations can draw on, each applying it to their own decision: closing a screen, pausing an event, timing a harvest, or warning a sector. The remainder of this report describes what has been built toward that capability, and who could use it.

# 2. The Solution: What Has Been Built

## a. In plain terms

Smart Patio Shield learns, from years of historical data, the conditions that produce wind-driven rain at a given location, and uses that learning to assess the risk for the hours at hand. Where it differs from an ordinary weather app comes down to two design choices.

**It is hyper-local and event-specific.** It predicts the specific damaging event (rain combined with gusting wind) for a particular location, not a general regional rain chance.

**It was built to test, not assume, what data is needed.** The project deliberately evaluated two very different ways of seeing the weather (ground-level measurements and live satellite imagery of the clouds) and measured how much each actually contributes rather than assuming more data is better. That test produced the project's central engineering finding, described in Section 2b.

## b. The role of the satellite: a decisive test

A reasonable expectation at the outset was that satellite imagery would meaningfully improve predictions, because it can show storms physically forming and approaching, whereas ground measurements infer them indirectly. The project built the satellite capability specifically to test that expectation. The result was clear and, commercially, valuable:

**Satellite imagery did not improve the predictions.** Rigorous testing showed that the satellite's genuine advantage is confined to heavier convective storms, while most of the damaging events in this setting are lighter rain that infrared imagery cannot see well. Combing the satellite with the weather data produced no improvement over the weather data alone. The deployed system therefore uses weather data only, a deliberate evidence-based decision that keeps it lightweight, cheaper to run, and simpler to license.

This is a stronger position than it may first appear. Rather than shipping a complex, expensive multimodal system on the assumption that more data must help, the project measured the contribution of each data source and kept only what earns its place. The result is a leaner product with lower running costs and a clear, honest story about why it is built the way it is.

## c. Three models, one honest question

The project built three predictive models, all answering the same question, "will this location experience wind-driven-rain event?", in three different ways: one using ground-level weather data, one using satellite imagery, and one fusing both. Building all three allowed a direct measurement of how much each data source contributes. That measurement is what revealed the weather-only model to be the right one to deploy, and it is the kind of disciplined comparison that underpins confidence in the final system.

# 3. What Has Been Achieved

## a. Current status

The project is a complete, validated working prototype. The full data pipeline is built and documented; all three models are trained and evaluated on held-test recent data; and a deployable inference service, with an interactive demonstration console, has been built and shown to work on live conditions. The design and engineering work is done; what remains is commercial validation, not further invention.

## b. Does it work? The evidence

The models were tested on recent data they had never seen during development, the true measure of whether a system will work in practice. The best model substantially outperformed both random chance and a simple "assume the weather continues" rule, and it did so consistently across the test period. In practical terms, when tuned to prioritise catching events, it identified the large majority of genuine wet-veranda hours. The balance between "catching every event" and "avoiding false alarms" is adjustable, and is properly a business decision: a hotel protecting expensive furnishings may accept more false alarms to miss fewer events, while an event organiser may prefer the reverse.

**A measured, honest result.** Performance was established using a strict evaluation that hols out recent data and never lets a model "peek" at the answers, the same discipline used in credible scientific work. The project also quantified how much of the model's skill comes from simple persistence versus genuine added insight, and confirmed the added insight is rail and concentrated where it matters most: anticipating rain that is just beginning, rather than merely noting rain already underway. The system was further stress-tested against Hurricane Melissa, which fell within the evaluation period, with results reported for both including and excluding that extreme weather event so that a single storm neither flatters or distorts the picture.

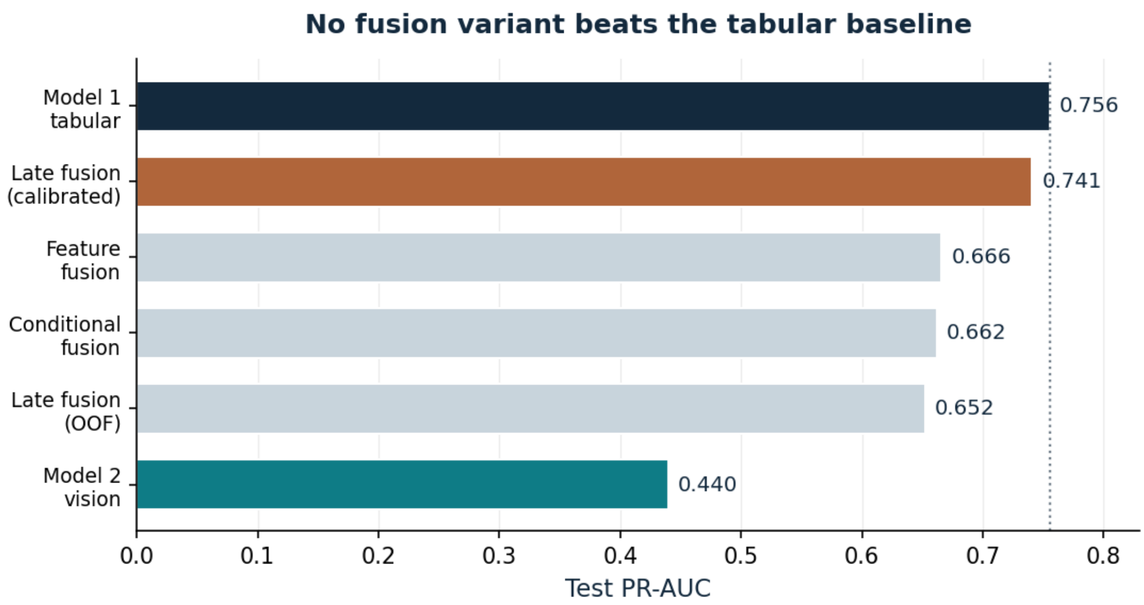


*Figure 1. How the model compared on the held-out test data. The weather-only model (top) leads; no combination with satellite imagery improved on it. This is the evidence behind the decision to deploy the lightweight weather-only system.*

## c. What is not yet done

In the interest of an honest account: the system currently covers two locations and would need a short, automated onboarding step for each new site; the precise definition of a "wet-veranda" event is reasonable but not yet calibrated against on-the-ground observations; a longer forecasting horizon (predicting further ahead, rather than assessing the current hour) is a natural next step but not yet delivered; and, most importantly for the commercial case, no paying customer has yet validated demand. None of these is a fundamental obstacle, but each is a real, named piece of remaining work.

# 4. Value Proposition and Business Model

## a. The product

The product is not a single gadget but a prediction service: a capability, delivered over the internet as an interface that other systems and organisations can draw on. A user, whether a building's automated screen, a hotel's operations dashboard, or a government agency's monitoring system, asks the service about a location, and receives a clear, current assessment of wind-driven-rain risk together with a recommended action.

## b. How it makes money

The intended model is licensing per location, on a recurring basis: a subscription for each protected site or monitored area. This suits the nature of the product well: the prediction capability is built once and can then serve many customers and locations, so each additional site adds revenue at low additional cost. The recurring structure also aligns the provider's incentive with the customer's: the service must keep working, every day, to retain the subscription.

## c. What protects the business

The defensible advantage is not the choice of algorithm, since those are widely available. It rests on hour things that are harder to copy:

- **Localised calibration.** The system is tuned to the specific microclimate and the specific definition of a damaging event for each site. Two locations only few kilometres apart can behave very differently, as this project demonstrated for Kingston and Montego Bay.

- **A lightweight, low-cost footprint.** Because the project proved satellite imagery is unnecessary, the service runs on free weather data and modest computing, with no satellite feed and no specialised hardware. That keeps the running costs lows and undercuts any competitor carrying the weight of a heavier system.

- **An automatable onboarding pipeline.** Because everything is built on free public data keyed to coordinates, adding a new location is largely automated process rather than a bespoke project.

- **Honest, validated performance.** The rigour behind the predictions, and the willingness to state plainly what the system can and cannot do, is itself a differentiator in a field crowded with vague claims.

# 5. Stakeholders and Market

The capability is general, but the value it creates is specific to each kind of user. The table below maps the most promising stakeholder groups, several of them named priorities for early engagement, to their particular exposure and the value the service would offer them.

| Stakeholder | Exposure to wind-driven rain | Value offered |
|---|---|---|
| Hotels & resorts | Outdoor dining, pools, beach events, weddings | Protect guest experience and assets; reduce disruption to high-value outdoor events |
| Restaurants | Open-air and patio seating | Protect seating and service; give staff timely, location-specific warnings |
| Agriculture (farms, RADA, Ministry of Agriculture) | Spraying and harvesting windows; sun-drying of coffee, pimento, cocoa | Time weather-sensitive operations; protect drying crops from sudden rain |
| Tourism agencies (Ministry of Tourism; JTB) | Sector-wide visitor experience; outdoor attractions | Sector-level decision support and visitor-facing warnings |
| Entertainment & events (Ministry; promoters) | Outdoor concerts, festivals, stage shows | Go / no-go and protective decisions for outdoor events |
| Screen & awning manufacturers | (Enabler) hardware lacks intelligence control | An embedded intelligence layer: the trigger that makes automated screens genuinely smart |

## a. Why these sectors, and why Jamaica

Tourism and agriculture are cornerstones of Jamaica's economy, and both are acutely weather-exposed in ways this system directly addresses: outdoor hospitality on one side, rain-sensitive cultivation and crop-drying on the other. Jamaica's strong outdoor-events and entertainment culture adds a third naturally exposed sector. The country's pronounce microclimates, the same rain-shadow contrast tis project measured between its two study cities, mean that hyper-local prediction is a genuine necessity rather than a marginal refinement, which strengthens the case for a localised service over generic forecasts.

## b. A clear-eyed view of demand

**Demand is hypothesised, not yet proven.** The stakeholder groups above have a plausible, specific, and in several cases acute exposure to the problem this system solves. That is a strong reason to believe demand exists. It is not, however, the same demonstrated willingness to pay. No customer interviews, letters of intent, or paid pilots have been conducted. The single most valuable next commercial step is therefore not more technology but direct engagement with a small number of these stakeholders to test the hypothesis, described in Section 8.

# 6. Cost and Resources

A notable feature of the project, and a direct commercial advantage, is how little it costs to run. The satellite finding has sharpened that advantage, since the deployed system needs neither satellite data nor specialised hardware.

| Resource | Source | Cost implication |
|---|---|---|
| Weather data | Public reanalysis archive (Open-Meteo) | Free; no licensing |
| Climate index | Public NOAA data | Free |
| Satellite imagery | Evaluated but not used in the deployed product | No ongoing cost |
| Computing | Commodity laptop; no GPU needed to serve | Minimal; pay-as-used |
| Effort | Single developer, ~6 month capstone | Concentrated, documented |

The strategic point is that the inputs are free and public, and the pipeline that turns them into predictions is automated and lightweight. The cost of serving an additional location is low and predictable. That is the foundation of a scalable service, and it lowers the barrier to early commercialisation.

# 7. Risks and Limitations

An honest assessment names the risks plainly, in two groups.

## a. Technical and delivery risks

- **Presented-focused prediction.** The current system is strongest at assessing the immediate situation rather than forecasting hours ahead; extending the forecast horizon is planned and is a natural next step, but is not yet delivered.

- **Light-rain detection ceiling.** The hardest events to call are the lightest ones. The project established this limit precisely, and showed that satellite imagery, the obvious candidate to help, does not resolve it. That sets honest expectations rather than leaving a hidden weakness.

- **Calibration to ground truth.** The definition of a damaging event is reasonable but not yet tuned against logged real-world observations.

## b. Commercial risks

- **Unproven demand.** As stated, willingness to pay has not been tested. This is the principal commercial risk and the focus of the recommended next steps.

- **Onboarding effort per site.** Although largely automated, each new location requires a calibration step; at scale, this must be made genuinely turnkey.

- **Sector procurement realities.** Government agencies, in particular, have procurement processes and timelines that a small provider must be prepared for.

# 8. Roadmap and Recommendations

## a. Strengthening the prototype (near-term)

1. Calibrate the wind-driven-rain event definition against a log of real on-the-ground observations, so the target reflects what users actually experience as damaging.
2. Extend the prediction horizon from the current hour to a short forecast ahead, increasing the lead time available to act.
3. Make site onboarding genuinely turnkey, so that adding a location is demonstrably low-cost and low-effort.

## b. Testing the business (commercial)

4. Engage a small number of named stakeholders, for example one hotel, one agricultural body, and one events organiser, to test the demand hypothesis directly.
5. Run one or two no-cost or low-cost pilots at real sites, calibrating the event definition against on-the-ground observation and gathering evidence of value.
6. On the strength of pilot evidence, approach sector bodies (tourism, agriculture, entertainment) ad screen manufacturers with a concrete, evidence proposition.

## c. The single most important recommendation

**Validate demand before scaling technology.** The technology is complete and de-risked, and the satellite finding has already stripped out unnecessary cost and complexity. The largest remaining uncertainty is commercial, not technical. The highest-value next action is therefore to put the working prototype in front of a few real stakeholders and learn, quickly and cheaply, whether the plausible demand is real demand. Everything else follows from that answer.

## 9. Conclusion

Smart Patio Shield began as an academic capstone and has produced a complete, validated, working prototype that predicts a specific, costly, and currently unserved weather event, wind-driven rain at a precise location, using only free public data and modest computing. Its design is deliberately honest: it claims only what it has demonstrated, quantifies the contribution of every component, and is explicitly about what remains to be done. That discipline produced its defining engineering result, that the lightweight weather-only system is the one worth deploying, which is as much as commercial asset as a scientific one.

That same discipline frames the commercial opportunity. The need is real and specific; the stakeholders in tourism, agriculture, entertainment, hospitality, and screen manufacturing are identifiable and, in Jamaica's microclimate-rich and weather-exposed economy, plausibly well-served by exactly this capability. What remains is to test the commercial hypothesis directly with the very stakeholders this report names. The work to date has made that test inexpensive to run and credible to present, which is, for a project at this stage, the position worth being in.